2 **(a)** State what is meant by *coherent* light waves.

.....

..... [1]

(b) Coherent light of wavelength 530 nm is incident normally on a double slit, as illustrated in Fig. 2.1.

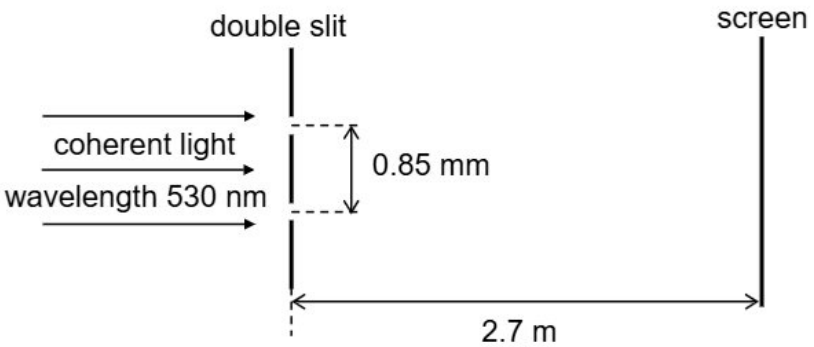


Fig. 2.1 (not to scale)

The separation of the two slits is 0.85 mm. A screen is placed parallel to the plane of the double slit at a distance of 2.7 m from the double slit.

Determine the separation of the fringes observed on the screen.

separation = ... m [2]

(c) Describe the changes to the appearance of the fringes when each of the following changes is made separately.

(i) The width of both slits is reduced without altering their separation.

.....

..... [1]

(ii) The screen is rotated so that it is no longer parallel to the plane of the two slits.

.....

.....

.....

.....

..... [2]

[Total: 6]

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